

Average rates of change



- 1
- a -4 b -3 c -7 d Variable

2 Constant

- 3
- a -1 b -1 c -1 d Linear

- 4
- a -4 b -4 c -4 d Linear

- 5
- a Constant b Variable c Constant d Variable
- e Variable f Variable g Variable h Variable

- 6
- a $\frac{23}{4}$ b 5 c $\frac{23}{5}$ d Non-linear

7 Non-linear

- 8
- a 1
- b 2
- c 4
- d

Interval	$[3, 4]$	$[4, 5]$	$[5, 6]$
Average rate of change	8	16	32

- 9
- a $\frac{9}{10}$
- b The cost of a call increases by \$0.90 for each additional minute.

10



a 500

b Aside from the pre-ordered tickets, there were 500 tickets purchased each day.

11

a $-\frac{1}{250}$

b The price of each toy decreases by \$1.00 for each additional 250 units in supply.

12

a 0

b The number of earthquakes per year is a constant value.

Applications

13

a $y = 11$

b $y = 26$

c $\frac{15}{4}$

14 272

15 \$300 per year

16 60

17 6.5 insects per week

18 1.6

19

a

i 9 ft/s

ii 3 ft/s

iii -3 ft/s

iv -9 ft/s

b Between $t = 0$ and $t = 1$, and between $t = 3$ and $t = 4$

c The ball is falling down.

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a

Year	0	1	2	3	4	5
Tobia's staff	310	269	228	187	146	105
Gwen's staff	310	302	286	262	230	190

- b -41
- c -32
- d Linear, because it decreases by the same amount each year.
- e Non-linear, because it decreases by more each year.
- f Tobias's
- g Gwen's

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a

Month	0	1	2	3	4	5
Mario's clients	0	27	54	81	108	135
Christa's clients	0	6	18	36	60	90

- b Linear
- c Non-linear
- d 27
- e 18
- f Mario's
- g Christa's

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a

Month	0	1	2	3	4	5
Value of Harry's investment	2000	2060	2120	2180	2240	2300
Value of Skye's investment	2000	2040	2080.8	2122.42	2164.87	2208.17

- b \$60
- c \$40.8
- d Linear
- e Non-linear



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a $y = 4x + 20$

c 4 turbines each year

e Option A

b $y = 20(1.10)^x$

d 9.59 turbines each year